14X023 Software Modeling and Verification **Assistant**: Matteo De Francesco
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Exercise Set 9 10 November 2025

1 Abstract Interpretation I

1.1 Bulding CFG

Given the following programs

```
sillyprogram(a,b) =
  while (a > 0) do
    if a mod 2 = 0 then
      a := a/b
    else
      b := b-1;
      a := a+1
    endif
  done
```

```
binarysearch(a, elem) =
  l := 0;
  u := a.length-1;
  while (l <= u) do
    m := (u-l)/2;
    if (a[m] < elem) then
      l := m+1
    else
      if (a[m]>elem) then
        u := m+1
      else
        m;
      endif;
    endif
  done;
  -1
```

- Draw the CFG of each program
- What are the possible behaviours of `sillyprogram` depending on the values of `a` and `b`?

1.2 Defining CFG

Define the function that builds a CFG, from a program P. To make the definition easier, we advise that the function also takes as inputs ip and rp which will be the entry and return control points of the resulting CFG. At last, we will define $CFG(P, ip, rp)$; it should depend on the grammar that defines the instructions of the language.

- Define $CFG(P, ip, rp)$ for every rule of the grammar that defines the instructions
- What is the initial call of CFG (to build the entire CFG of the program)?

1.3 Collecting semantics

- Write the system of equations representing the collecting semantics of `sillyprogram`.
- Solve the system of equations for $a = 5$ and $b = 3$ at the beginning of the execution.
- What can you conclude, based on the values of V, about the execution of the program? (Recall, $V(\alpha)(x)$ is the set of all values variable x has been set to during the execution of the program at the Control Point α)

Now, instead of having, for all α, x, $V(\alpha)(x)$ represented as a set of values, we can abstract this set by using intervals (for example, if $V(\alpha)(x) = \{2, 3, 4, 5, 6\}$, then we abstract it as $V(\alpha)(x) = [2, 6]$, if $V(\alpha)(x) = \{\ldots, -3, -2\}$, then we abstract it as $V(\alpha)(x) = (-\infty, -2])$. For each variable, its set of values is then abstracted away by an interval in $[a, b], (-\infty, a], [a, +\infty)$, $(-\infty, +\infty)$ for some integer values of a and b.

- Solve again the system of equations of the previous exercise by using this abstraction.

We can restrict even more the encoding, by just remembering the *sign* of the variable. For each variable, its set of values is now abstracted by one of the following elements ≤ 0, ≥ 0, < 0, > 0, $= 0$, $\neq 0$, $\perp$ (no value), $\top$ (all values are possible).

- Solve again the system by using this abstraction. What can you observe?